

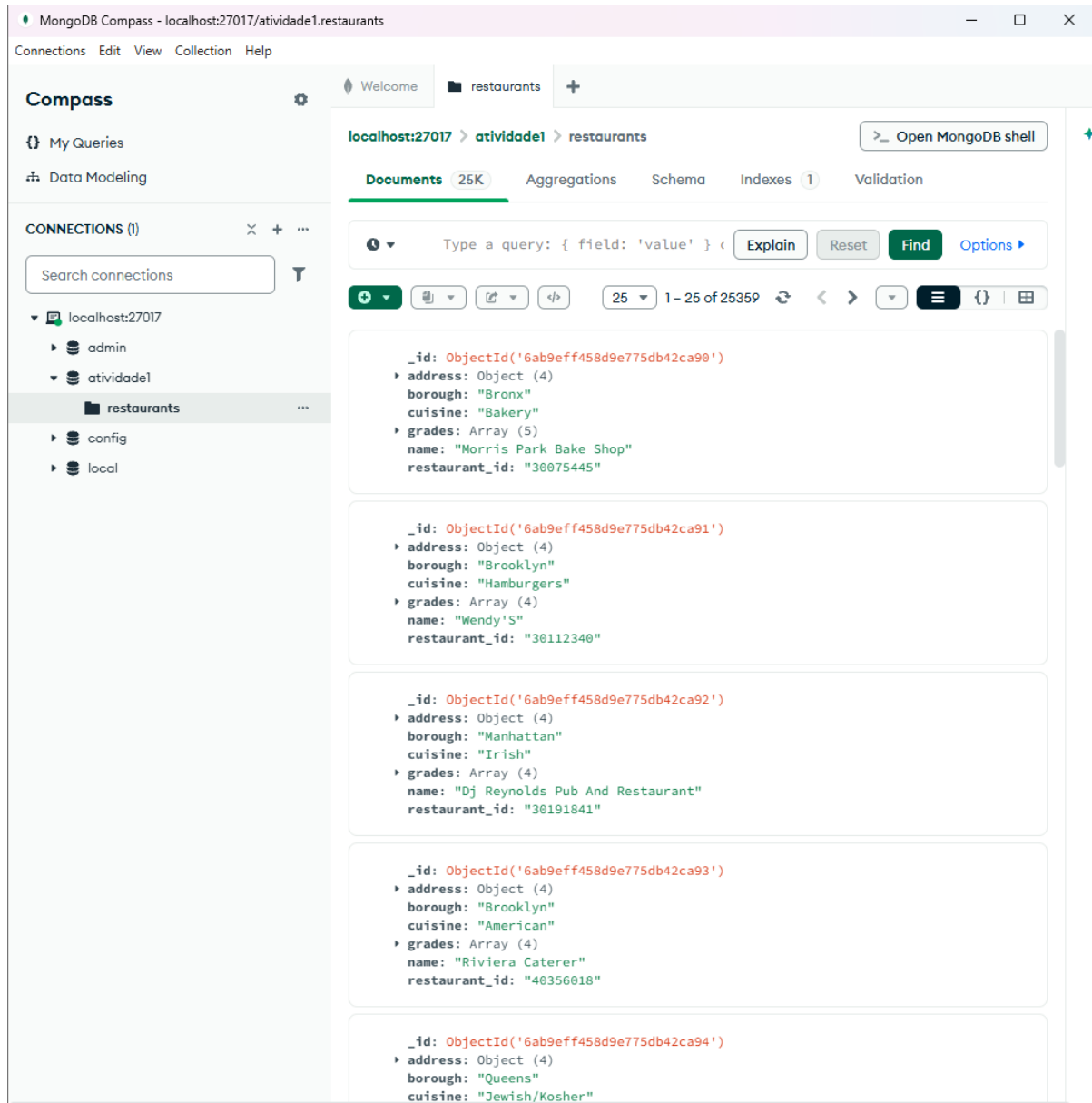
DOCUMENTO ATIVIDADE 1 – MONGODB

Implementação e Gerenciamento de Bancos de Dados NoSQL

Código no GitHub: github.com/joseniltoncirne/Implementacao-e-Geren-de-banco-de-dados/tree/main/individuais/atividade01-mongodb

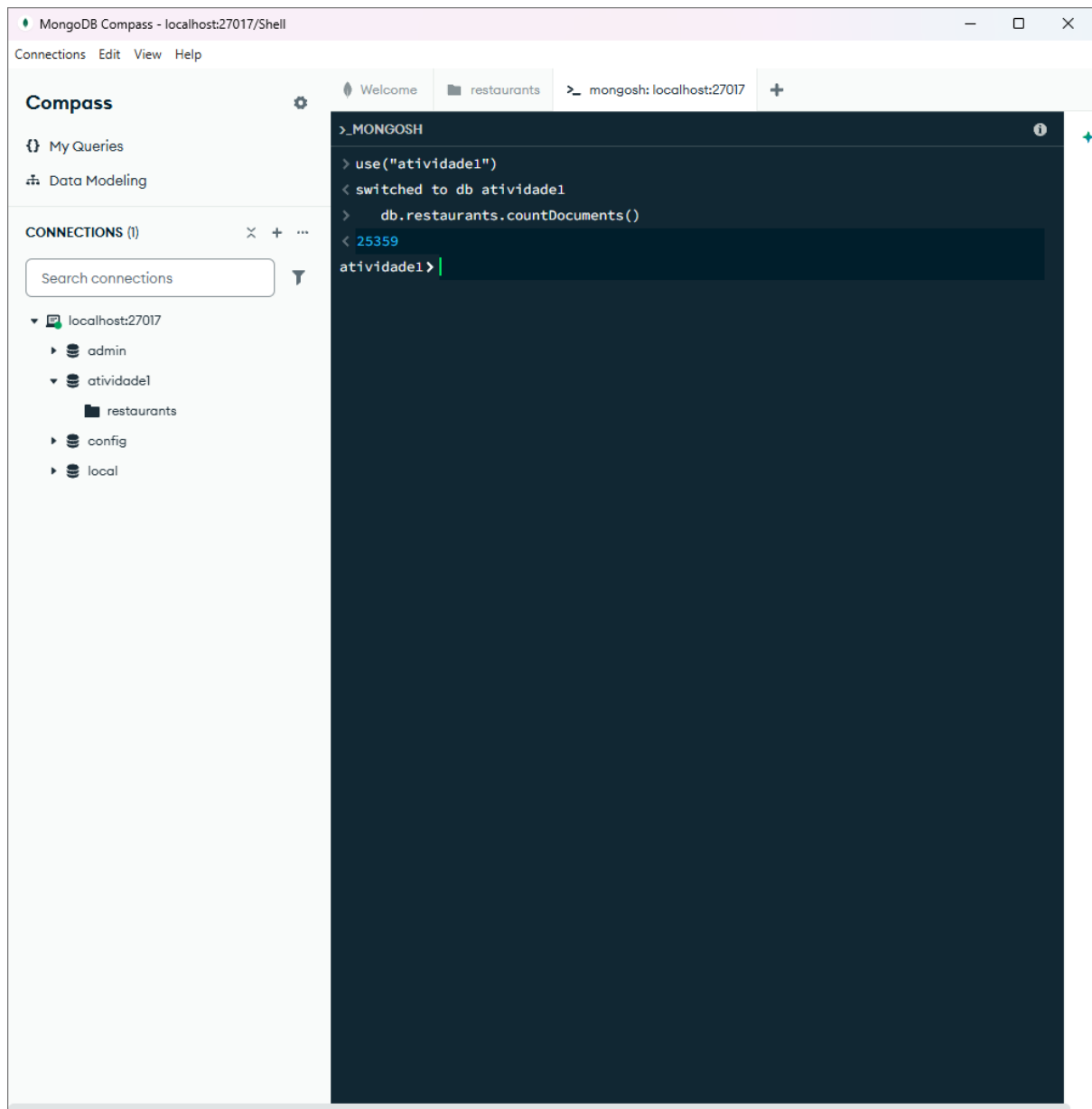
ETAPA 1: Importando os dados

Importei o arquivo restaurantes.json pelo MongoDB Compass (botão Import data), no banco atividade1, coleção restaurants:



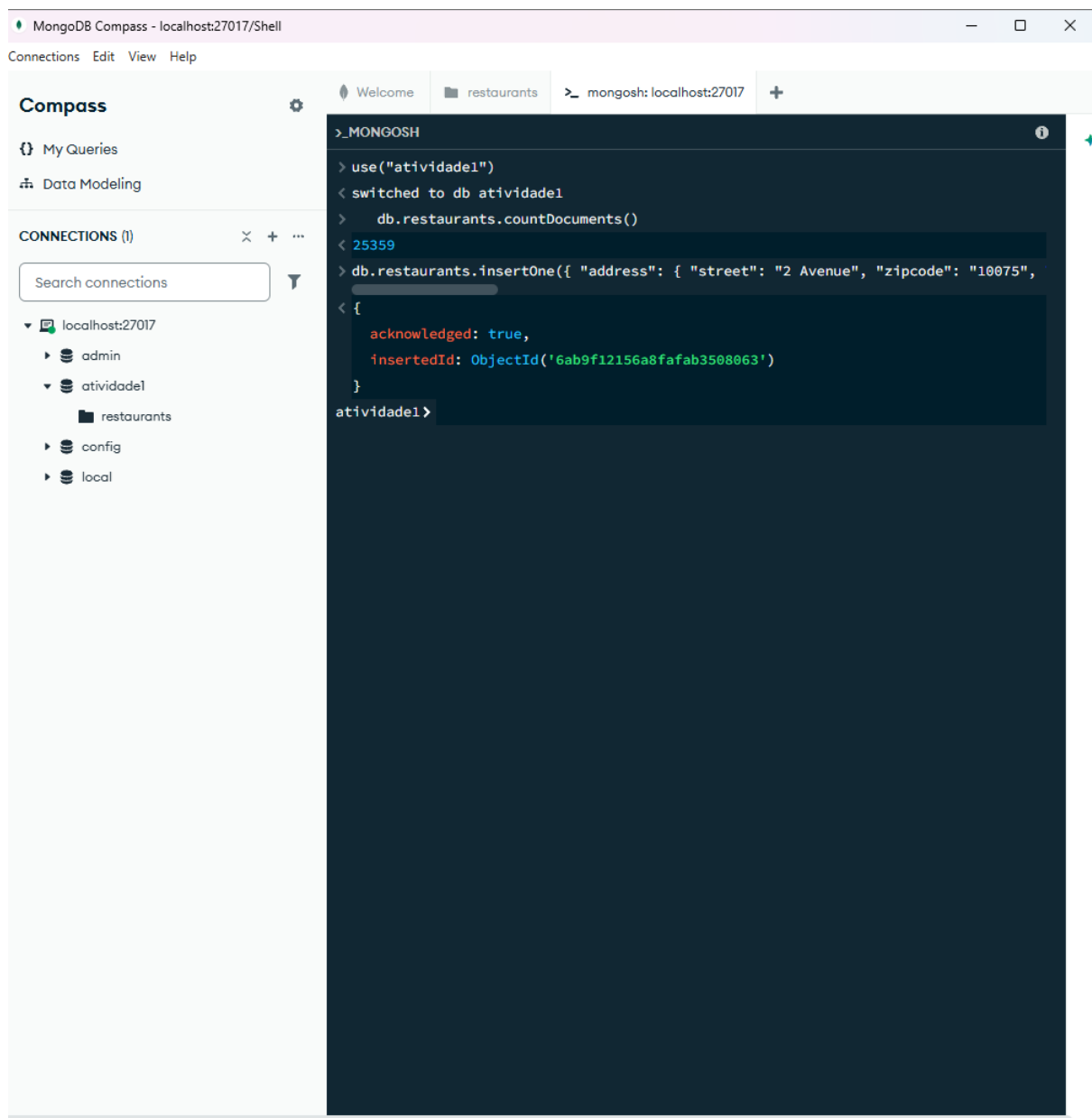
Conferindo no mongosh se os dados entraram:

```
use atividade1  
db.restaurants.countDocuments()
```



ETAPA 2: Adicionando um restaurante novo

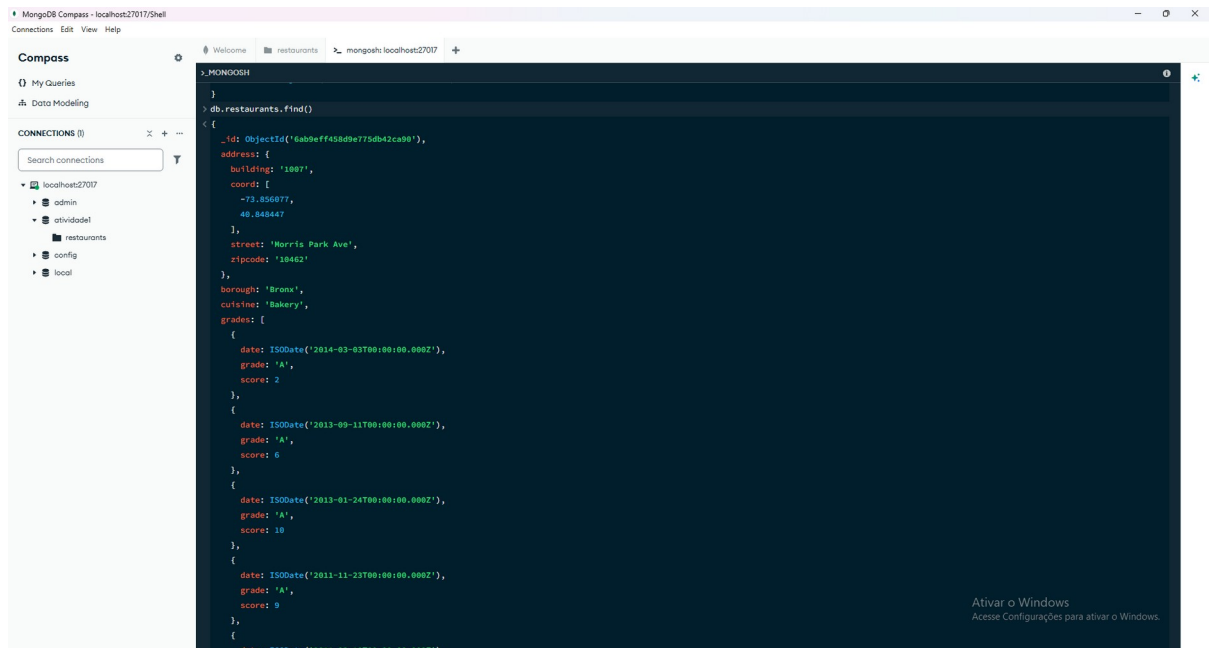
```
db.restaurants.insertOne({
  "address": {
    "street": "2 Avenue",
    "zipcode": "10075",
    "building": "1480",
    "coord": [-73.9557413, 40.7720266]
  },
  "borough": "Manhattan",
  "cuisine": "Italian",
  "grades": [
    { "date": ISODate("2014-10-01T00:00:00Z"), "grade": "A", "score": 11 },
    { "date": ISODate("2014-01-16T00:00:00Z"), "grade": "B", "score": 17 }
  ],
  "name": "Vella",
  "restaurant_id": "41704620"
})
```



ETAPA 3: Fazendo buscas

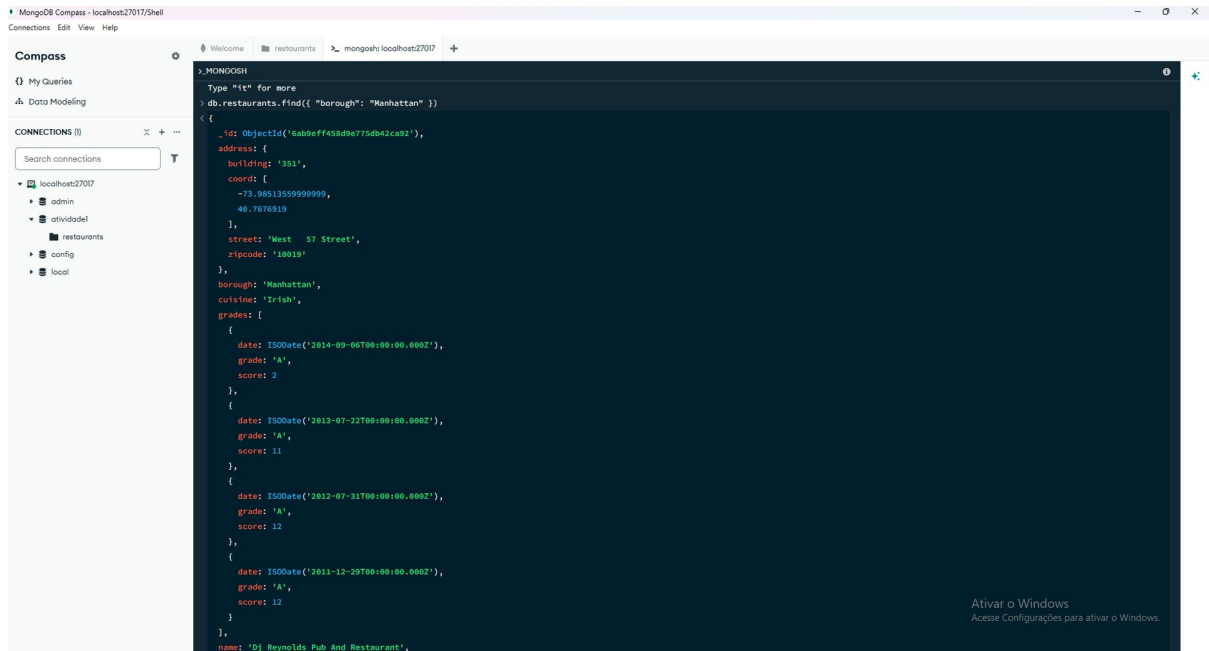
a) Mostrar todos os restaurantes

```
db.restaurants.find()
```



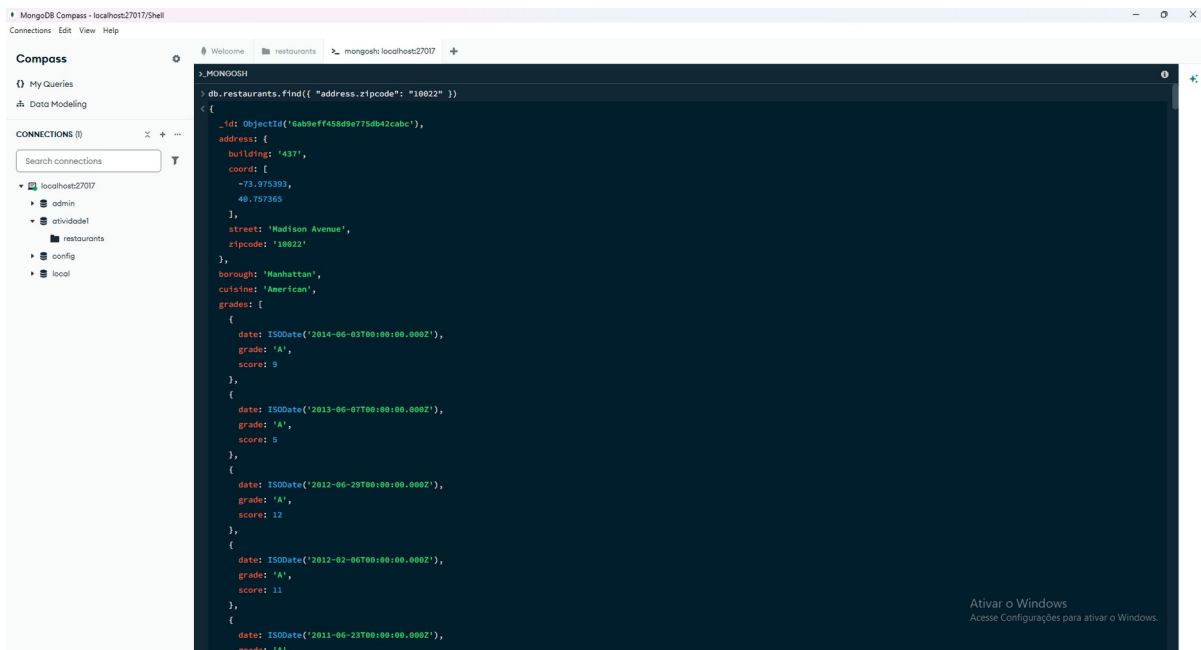
b) Restaurantes de Manhattan

```
db.restaurants.find({ "borough": "Manhattan" })
```



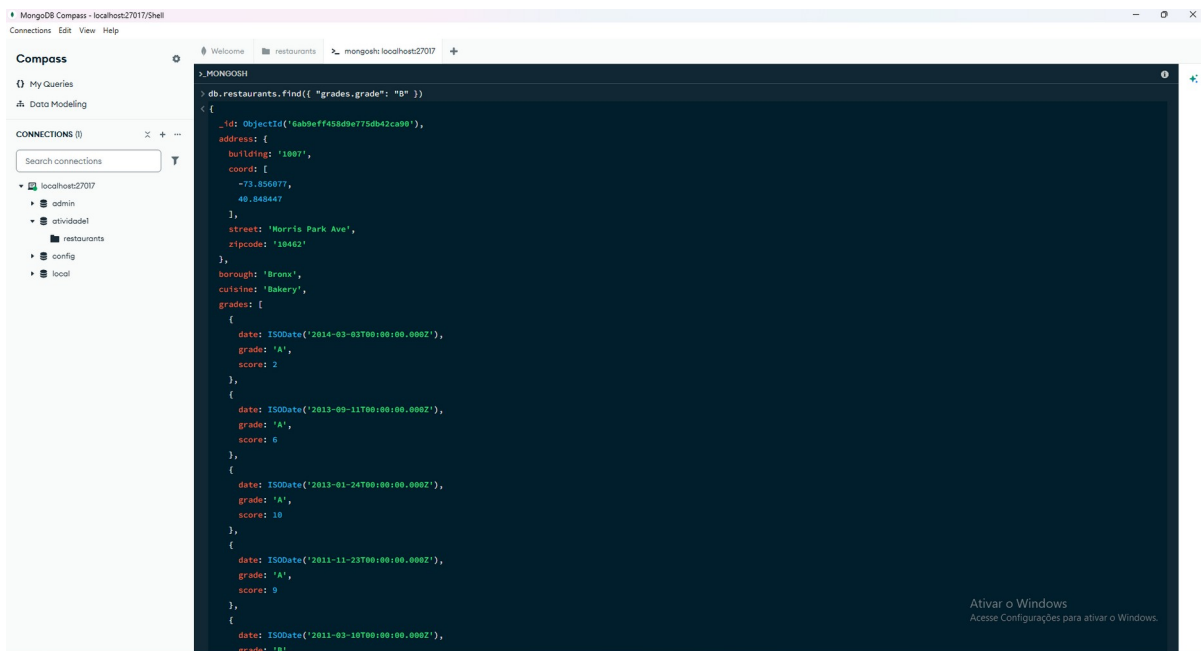
c) Restaurantes com CEP 10022

```
db.restaurants.find({ "address.zipcode": "10022" })
```



d) Restaurantes que já tiraram nota B

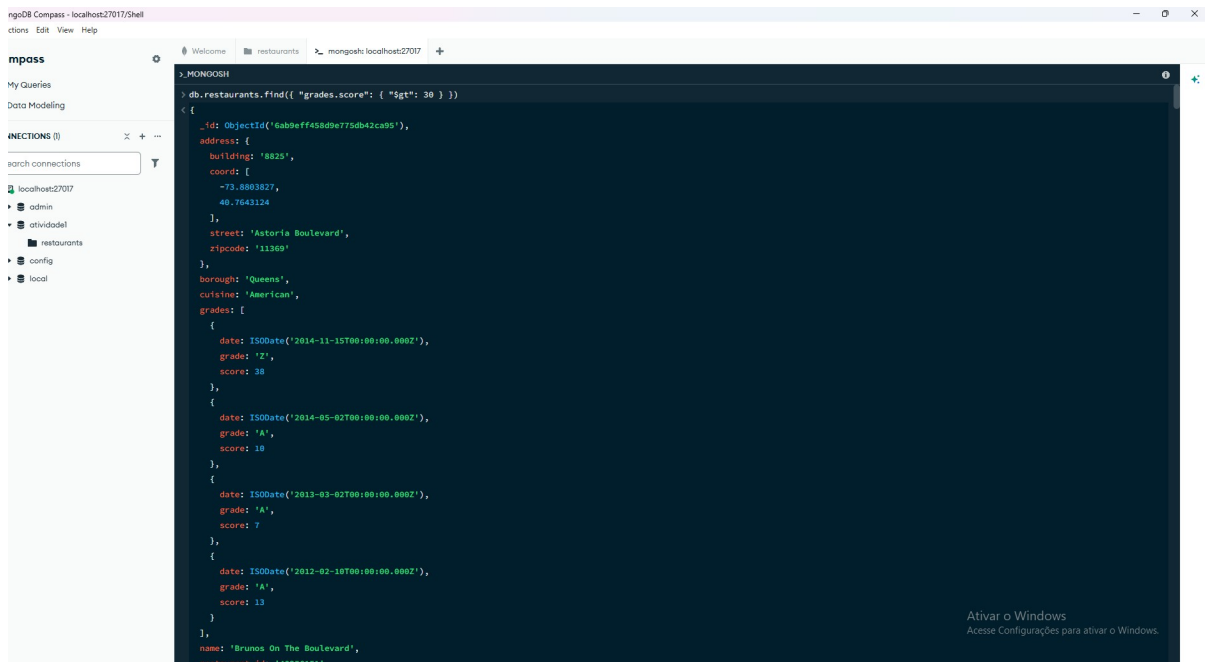
```
db.restaurants.find({ "grades.grade": "B" })
```



ETAPA 4: Buscas com filtros

a) Restaurantes com pontuação acima de 30

```
db.restaurants.find({ "grades.score": { "$gt": 30 } })
```

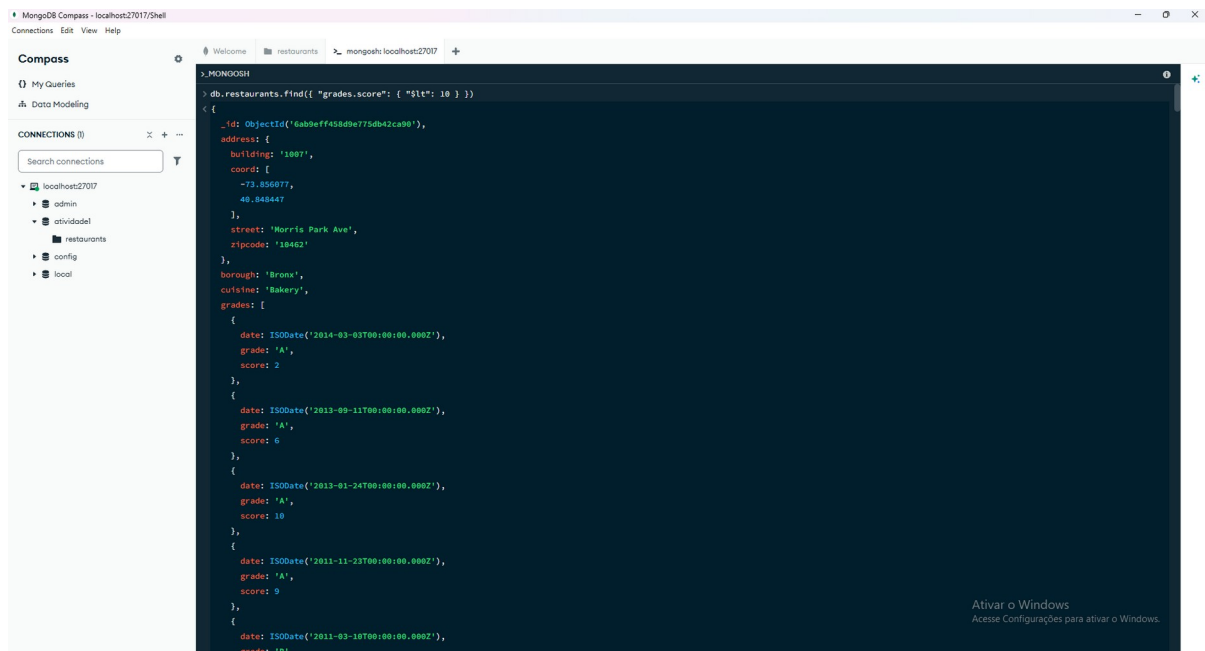


The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'localhost:27017' selected. The main panel shows the 'restaurants' collection. The query bar contains the command: `db.restaurants.find({ "grades.score": { "$gt": 30 } })`. The results pane shows a single document for a restaurant named 'Brunos On The Boulevard' with a score of 38. The document structure includes fields for _id, address (building, coord, street, zipcode), borough, cuisine, grades (an array of objects with date, grade, and score), and name.

```
> db.restaurants.find({ "grades.score": { "$gt": 30 } })
< {
  _id: ObjectId('6ab9eff43d9e775db42ca95'),
  address: {
    building: '8835',
    coord: [
      -73.8803827,
      40.7643124
    ],
    street: 'Astoria Boulevard',
    zipcode: '11369'
  },
  borough: 'Queens',
  cuisine: 'American',
  grades: [
    {
      date: ISODate('2014-11-15T00:00:00.000Z'),
      grade: 'Z',
      score: 38
    },
    {
      date: ISODate('2014-05-02T00:00:00.000Z'),
      grade: 'A',
      score: 18
    },
    {
      date: ISODate('2013-03-02T00:00:00.000Z'),
      grade: 'A',
      score: 7
    },
    {
      date: ISODate('2012-02-18T00:00:00.000Z'),
      grade: 'A',
      score: 13
    }
  ],
  name: 'Brunos On The Boulevard',
  restaurant_id: '48956191'
}
```

b) Restaurantes com pontuação abaixo de 10

```
db.restaurants.find({ "grades.score": { "$lt": 10 } })
```

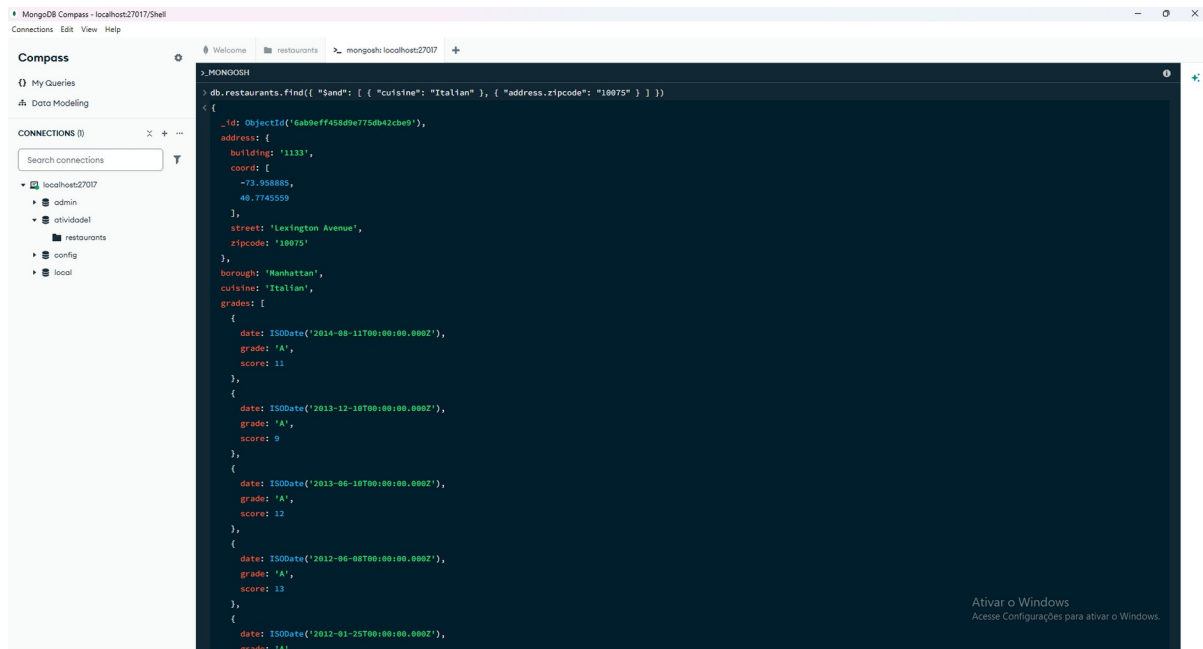


The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'localhost:27017' selected. The main panel shows the 'restaurants' collection. The query bar contains the command: `db.restaurants.find({ "grades.score": { "$lt": 10 } })`. The results pane shows a single document for a restaurant named 'Morris Park Ave' with a score of 2. The document structure includes fields for _id, address (building, coord, street, zipcode), borough, cuisine, grades (an array of objects with date, grade, and score), and name.

```
> db.restaurants.find({ "grades.score": { "$lt": 10 } })
< {
  _id: ObjectId('6ab9eff43d9e775db42ca96'),
  address: {
    building: '1887',
    coord: [
      -73.856077,
      40.848447
    ],
    street: 'Morris Park Ave',
    zipcode: '10462'
  },
  borough: 'Bronx',
  cuisine: 'Bakery',
  grades: [
    {
      date: ISODate('2014-03-03T00:00:00.000Z'),
      grade: 'A',
      score: 2
    },
    {
      date: ISODate('2013-09-13T00:00:00.000Z'),
      grade: 'A',
      score: 6
    },
    {
      date: ISODate('2013-01-24T00:00:00.000Z'),
      grade: 'A',
      score: 18
    },
    {
      date: ISODate('2011-11-23T00:00:00.000Z'),
      grade: 'A',
      score: 9
    },
    {
      date: ISODate('2011-03-18T00:00:00.000Z'),
      grade: 'B',
      score: 1
    }
  ],
  name: 'Morris Park Ave',
  restaurant_id: '48956192'
}
```

c) Restaurantes italianos que ficam no CEP 10075

```
db.restaurants.find({
  "$and": [
    { "cuisine": "Italian" },
    { "address.zipcode": "10075" }
  ]
})
```



d) Restaurantes italianos ou que ficam no CEP 10075

```
db.restaurants.find({
  "$or": [
    { "cuisine": "Italian" },
    { "address.zipcode": "10075" }
  ]
})
```

